# User Stories

Agile Team: Hillary Itlhabanyeng, Inam Klaas, Onika Gobane, Siphesihle Khinase, Sesethu Nciti, Nondumiso Ngobese, Liyabona Ngece

**Note on team changes:** Hillary Itlhabanyeng and Inam Klaas left the group during the project. Nondumiso Ngobese and Liyabona Ngece joined the group to replace them and took over a share of the work, which is why their names appear on the later sprints.

## Sprint 1 – Team Setup and Planning

### Day 1: Team Formation and Project Setup

**User Story:** As a project team member, I want to be assigned a role, join the team's communication platforms, and gain access to the project repository so that I can collaborate effectively with the team throughout the project.

**Tasks**

- [x] ~~Form project group – Siphesihle~~
- [x] ~~Exchange contact details~~
- [x] ~~Create GitHub repository – Onika Gobane~~
- [x] ~~Create WhatsApp group – Sesethu Nciti~~

### Day 2: Roles and Responsibilities

**User Story:** As a project team member, I want to have a clearly defined role and responsibilities so that I know what tasks I am responsible for and can contribute effectively to the development of the ParkLink application.

**Tasks**

- [x] ~~Assign Team Leader – Onika Gobane~~
- [x] ~~Assign Researcher – Sesethu Nciti~~
- [x] ~~Assign UI Designer – Siphesihle~~
- [x] ~~Confirm responsibilities with all team members~~

### Day 3: Team Charter

**User Story:** As a project team member, I want to establish a team charter with clear goals, communication methods, and team rules so that the team can work together effectively and successfully deliver the ParkLink application.

**Tasks**

- [x] ~~Define project vision~~
- [x] ~~Define project objectives~~
- [x] ~~Establish team rules~~
- [x] ~~Define communication methods~~

### Day 4: Communication Plan

**User Story:** As a project team member, I want to have a clear communication plan so that I can stay informed, collaborate effectively with the team, and receive project updates on time.

**Tasks**

- [x] ~~Select communication tools~~
- [x] ~~Create meeting schedule~~
- [x] ~~Define reporting procedures~~
- [x] ~~Define response times~~

## Sprint 2 – Community Research and Problem Identification

### Day 1: Community Research

**User Story:** As a commuter or vehicle owner, I want a parking application that understands the challenges faced by my community so that I can easily find available parking spaces and improve my parking experience.

**Tasks**

- [x] ~~Identify target community~~
- [x] ~~Research demographics~~
- [x] ~~Identify vehicle owners~~
- [x] ~~Identify common parking problems~~

### Day 2: Problem Identification

**User Story:** As a driver or commuter, I want a solution that addresses the main parking challenges in my community so that I can find safe, convenient, and available parking spaces quickly and efficiently.

**Tasks**

- [x] ~~Brainstorm community problems~~

- [x] ~~Select the most suitable problem~~
- [x] ~~Identify the target users~~
- [x] ~~Identify the causes of the problem~~

### Day 3: Surveys

**User Story:** As a potential ParkLink user, I want to share my parking experiences and challenges through a survey so that the development team can create a solution that meets the needs of the community.

**Tasks**

- [x] ~~Create Google Form~~
- [x] ~~Share survey~~
- [x] ~~Monitor survey responses~~
- [x] ~~Analyse survey results~~

### Day 4: Data Gathering

**User Story:** As a project team member, I want to collect and organise data from users so that I can understand their parking challenges and use the information to develop an effective ParkLink solution.

**Tasks**

- [x] ~~Conduct interviews~~
- [x] ~~Collect survey responses~~
- [x] ~~Record responses~~
- [x] ~~Organise collected data~~

## Sprint 3 – Business Case and Feasibility

### Day 1: Business Case

**User Story:** As a project stakeholder, I want a well-developed business case so that I can understand the value, feasibility, costs, and expected benefits of the ParkLink application before development proceeds.

**Tasks**

- [x] ~~Define business objectives~~
- [x] ~~Conduct market analysis~~
- [x] ~~Develop revenue model~~

- [x] ~~Perform cost analysis~~

## Day 2: Value Proposition

**User Story:** As a driver or commuter, I want a parking solution that solves my parking challenges and provides convenient, secure, and reliable parking services so that I can save time and improve my overall parking experience.

**Tasks**

- [x] ~~Identify target customers~~
- [x] ~~Define customer problems~~
- [x] ~~Define the ParkLink solution~~
- [x] ~~Define customer benefits~~

## Day 3: Feasibility Analysis

**User Story:** As a project stakeholder, I want the ParkLink project to be evaluated for technical, financial, operational, and scheduling feasibility so that I can determine whether the project is practical, achievable, and worth implementing.

**Tasks**

- [x] ~~Assess technical feasibility~~
- [x] ~~Conduct economic feasibility analysis~~
- [x] ~~Evaluate operational feasibility~~
- [x] ~~Review schedule feasibility~~

## Day 4: Start-ups Cost Research

**User Story:** As a project stakeholder, I want to understand the estimated start-up costs of the ParkLink application so that I can plan the required budget and ensure the project is financially viable.

**Tasks**

- [x] ~~Research domain registration costs~~
- [x] ~~Research web hosting costs~~
- [x] ~~Estimate marketing costs~~
- [x] ~~Calculate the total start-up budget~~

# Sprint 4 – Requirements and Risk Management

## Day 1: Project Requirements

**User Story:** As a stakeholder, I want the ParkLink application requirements to be clearly defined and documented so that the development team can build a system that meets user needs and business objectives.

**Tasks**

- [x] ~~Gather stakeholder requirements~~
- [x] ~~Identify functional requirements~~
- [x] ~~Identify non-functional requirements~~
- [x] ~~Prioritise requirements~~

## Day 2: Resources Requirements

**User Story:** As a project manager, I want to identify the software, hardware, human resources, and budget required for the ParkLink project so that the team has everything needed to develop and deliver the application successfully.

**Tasks**

- [x] ~~Identify software requirements~~
- [x] ~~Identify hardware requirements~~
- [x] ~~Identify human resources~~
- [x] ~~Estimate project budget~~

## Day 3: Risk Analysis

**User Story:** As a project manager, I want to identify and assess potential project risks so that I can minimise their impact and ensure the successful delivery of the ParkLink application.

**Tasks**

- [x] ~~Identify project risks~~
- [x] ~~Assess the likelihood of each risk~~
- [x] ~~Assess the impact of each risk~~
- [x] ~~Develop risk mitigation strategies~~

### Day 4: Risk Logs

**User Story:** As a project manager, I want to maintain a risk log so that project risks are tracked, monitored, and managed effectively throughout the development of the ParkLink application.

**Tasks**

- [x] ~~Create risk register~~
- [x] ~~Record identified risks~~
- [x] ~~Assign risk owners~~
- [x] ~~Review and update risks~~

## Sprint 5 – Product Backlog and MVP

### Day 1: User Stories

**User Story:** As a product owner, I want to create and prioritise user stories so that the development team can understand user needs and deliver valuable features for the ParkLink application in each sprint.

**Tasks**

- [x] ~~Create buyer user stories~~
- [x] ~~Create seller user stories~~
- [x] ~~Prioritise user stories~~
- [x] ~~Estimate story points~~

### Day 2: Product Backlog

**User Story:** As a Product Owner, I want to create and prioritise a product backlog so that the development team can focus on delivering the most valuable features for the ParkLink application in each sprint.

**Tasks**

- [x] ~~Create the product backlog~~
- [x] ~~Prioritise backlog items~~
- [x] ~~Estimate effort for each backlog item~~
- [x] ~~Assign priorities~~

### Day 3: MVP Planning

**User Story:** As a Product Owner, I want to define the Minimum Viable Product (MVP) so that the development team can deliver the most important features first and release a functional version of the ParkLink application quickly.

**Tasks**

- [x] ~~Define MVP features~~
- [x] ~~Select core functionality~~
- [x] ~~Define MVP objectives~~
- [x] ~~Review MVP scope~~

### Day 4: MVP Development

**User Story:** As a user, I want a functional MVP of the ParkLink application so that I can register, log in, and use the core parking features while the team gathers feedback for future improvements.

**Tasks**

- [x] ~~Complete discovery and scoping~~
- [x] ~~Identify the core user flow~~
- [x] ~~Implement the authentication process~~
- [x] ~~Define success metrics~~

## Sprint 6 – Wireframes

### Day 1: Wireframe Planning

**User Story:** As a user, I want the application to have a clear and intuitive layout so that I can easily navigate and find the features I need.

**Tasks**

- [x] ~~Identify required pages – Onika~~
- [x] ~~Sketch layouts – Onika~~
- [x] ~~Plan navigation – Siphesihle~~
- [x] ~~Review wireframe ideas – Sesethu~~

## Day 2: Wireframes

**User Story:** As a ParkLink user, I want every page of the website to have a clear, consistent, and easy-to-use layout so that I can browse vehicle parts, register an account, log in, manage my profile, search for products, add items to my cart, complete purchases, and access all website features without confusion. The wireframes should represent the complete user journey and ensure a smooth experience across the entire platform.

**Tasks**

- [x] ~~Design core website pages (Settings, Home, Login, Register) - Onika~~
- [x] ~~Design marketplace pages (Product Listings, Product Details, Cart, Checkout) - Sesethu~~
- [x] ~~Design user account pages (Profile, Landing, Forgot Password, Contact Us, About Us) - Siphesihle~~
- [x] ~~Review, refine, and approve all wireframes - Onika, Sesethu & Siphesihle~~

## Day 3: Wireframes

**User Story:** As an administrator, I want a well-organized dashboard that allows me to manage users, products, and listings efficiently. As a customer, I want product listing pages that are easy to browse, search, and filter so that I can quickly find the vehicle parts I need. The wireframes should provide a clear structure for both administrative and customer-facing features before development begins.

**Tasks**

- [x] ~~Design Admin Dashboard wireframe - Onika~~
- [x] ~~Design Product Listing page wireframe - Sesethu~~
- [x] ~~Design Product Details and Search/Filter wireframes - Siphesihle~~
- [x] ~~Review and finalize all wireframes - Onika, Sesethu & Siphesihle~~

## Day 4: Wireframes

**User Story:** As a ParkLink user, I want to manage my account settings, update my personal information, and reset my password through a simple and secure interface so that I can keep my account up to date and regain access whenever needed. The wireframes should provide an intuitive layout that makes these tasks quick and easy to complete.

**Tasks**

- [x] ~~Design Settings page wireframe - Onika~~
- [x] ~~Design Password Reset page wireframe - Sesethu~~
- [x] ~~Design User Profile and Account Management wireframes - Siphesihle~~

- [x] ~~Review, refine, and approve all wireframes – Onika, Sesethu & Siphesihle~~

# Sprint 7 – Development

## Day 1: Front-End Development

**User Story:** As a user, I want a clear and responsive interface so that I can easily navigate the application and access the parking features I need.

**Tasks**

- [x] ~~Build core website pages (Home, Login, Register, Settings) – Onika~~
- [x] ~~Build marketplace pages (Product Listings, Product Details, Cart, Checkout) – Sesethu and Nondumiso~~
- [x] ~~Build user account pages (Profile, Landing, Forgot Password, Contact Us, About Us) – Siphesihle and Liyabona~~
- [x] ~~Review, test responsiveness, and approve all front-end pages – Onika, Sesethu, Nondumiso, Liyabona & Siphesihle~~

## Day 2: Database Design

**User Story:** As a developer, I want a well-structured database so that the application can store, retrieve and manage user, parking and booking data reliably.

**Tasks**

- [x] ~~Identify entities and attributes (Users, Parking Spaces, Bookings) – Onika~~
- [x] ~~Create the entity relationship diagram (ERD) – Sesethu~~
- [x] ~~Create user and account tables with primary and foreign keys – Siphesihle~~
- [x] ~~Create parking space and booking tables with relationships – Nondumiso~~
- [x] ~~Prepare sample data and document the database structure – Liyabona~~
- [x] ~~Review, refine, and approve the database design – Onika, Sesethu, Siphesihle, Nondumiso & Liyabona~~

## Day 3: Develop Java Backend (IntelliJ IDEA)

**User Story:** As a developer, I want a functional backend so that the application can process requests, apply business logic and communicate with the database.

**Tasks**

- [x] ~~Set up the Java project structure in IntelliJ IDEA – Sesethu~~
- [x] ~~Create models and database connection – Nondumiso & Onika~~

- [x] ~~Develop user and authentication services – Liyabona~~
- [x] ~~Develop parking and booking services and controllers – Siphesihle & Sesethu~~
- [x] ~~Review, test, and approve the backend code – Nondumiso, Siphesihle, Liyabona, Onika & Sesethu~~

### Day 4: Connect Frontend to Backend (API Integration)

**User Story:** As a user, I want the application's pages to display and save real data so that I can use the system's features, such as viewing and booking parking, from start to finish.

**Tasks**

- [x] ~~Create and document the API endpoints – Siphesihle & Liyabona~~
- [x] ~~Connect login, register, and account pages to the API – Onika~~
- [x] ~~Connect marketplace, cart, and checkout pages to the API – Sesethu & Nondumiso~~
- [x] ~~Test data flow, fix errors, and approve the integration – Liyabona, Sesethu, Onika, Nondumiso & Siphesihle~~

## Sprint 8 – Authentication, Testing and Risk Review

### Day 1: Authentication

**User Story:** As a PartLink user, I want to register and log in securely so that I can access the marketplace and connect with verified car parts sellers with confidence that my information is protected.

**Tasks**

- [x] ~~Plan the authentication flow (register, login, logout)~~
- [x] ~~Build the registration and login screens~~
- [x] ~~Test with different user types and record the results~~
- [x] ~~Fix defects and confirm the feature meets the Definition of Done~~

### Day 2: User Testing Requirements

**User Story:** As a PartLink team member, I want to define clear user testing requirements so that we can check the application works the way our users need it to and fix problems before final delivery.

**Tasks**

- [x] ~~Review the user stories and list the features to be tested~~
- [x] ~~Identify the user groups and test participants~~

- [x] ~~Write test scenarios and success criteria~~
- [x] ~~Prepare a feedback form and a method for recording test results~~

### Day 3: Updated Risk Analysis

**User Story:** As a PartLink team member, I want to update our risk analysis so that we can identify new risks early and take action to reduce their impact before final delivery.

**Tasks**

- [x] ~~Review the original risks and risk logs from Sprint 4~~
- [x] ~~Identify new risks found during development and testing~~
- [x] ~~Rate each risk by likelihood and impact and assign an owner~~
- [x] ~~Update the mitigation plan and risk log~~

### Day 4: System Testing

**User Story:** As a PartLink user, I want the application to work correctly from registration to payment so that I can find and buy verified car parts without errors or delays.

**Tasks**

- [x] ~~Run the test scenarios for all main features (authentication, payments, search and listings)~~
- [x] ~~Record the test results and log any defects~~
- [x] ~~Fix defects and re-test the affected features~~
- [x] ~~Confirm the results against the Definition of Done and sign off~~

## Sprint 9 – Feedback and Closure

### Day 1: Updated Risk Analysis

Final review and update of the project risk register to ensure all risks encountered during the development lifecycle are documented and mitigated before project closure.

**User Story:** As a project manager, I want to update the risk analysis and risk log so that all new and previously identified risks are assessed, tracked, and have mitigation strategies for the final delivery of the application.

**Tasks**

- [x] ~~Review previous risk register from Sprint 4~~
- [x] ~~Identify new risks encountered during development~~
- [x] ~~Assess likelihood and impact of new risks~~

- [x] ~~Finalise updated risk log document~~

## Day 2: Collect Feedback

Collecting user and stakeholder feedback on the developed ParkLink application prototype/MVP to validate usability and identify final improvements.

**User Story:** As a product owner, I want to collect and organize feedback from potential users so that the team can validate the solution and identify final improvements needed.

**Tasks**

- [x] ~~Create feedback form/Google form~~
- [x] ~~Conduct user testing sessions~~
- [x] ~~Share application prototype with targeted users~~
- [x] ~~Analyse feedback results and organise it accordingly~~

## Day 3: Final UI Improvements

Implementing final UI/UX improvements based on collected user feedback to ensure the ParkLink application is intuitive, consistent, and user-friendly.

**User Story:** As a ParkLink user, I want the application to have a polished, consistent, and easy-to-use interface so that I can navigate, search for parking, and use all features without confusion or errors.

**Tasks**

- [x] ~~Review feedback related to UI/UX~~
- [x] ~~Improve responsiveness and layout~~
- [x] ~~Test final UI across different devices~~
- [x] ~~Review, refine, and approve final UI~~

## Day 4: Sprint Retrospective

Conducting the final sprint retrospective and project closure meeting to reflect on the entire development process, document lessons learned, and finalize the project.

**User Story:** As a project team member, I want to conduct a sprint retrospective so that the team can reflect on what went well, what did not go well, and document lessons learned for future projects and final submission.

**Tasks**

- [x] ~~Schedule final team retrospective meeting~~
- [x] ~~Discuss what went well during the project~~

- [x] ~~Discuss challenges and what can be improved~~
- [x] ~~Finalize project documentation and presentation~~

# Sprint 10 – Testing & Final Delivery

## Day 1: Final Documentation

**User Story:** As a project team member, I want to compile and finalize all project documentation so that the complete ParkLink application development process is well-documented and ready for final submission.

**Tasks**

- [x] ~~Compile all user stories and product backlog~~
- [x] ~~Finalize business case and feasibility documents~~
- [x] ~~Proofread and format final documentation~~
- [x] ~~Submit final documentation for review~~

## Day 2: Final Project Presentation & Demonstration

**User Story:** As a stakeholder, I want to see a final demonstration of the ParkLink application so that I can understand the complete functionality and value delivered.

**Tasks**

- [x] ~~Prepare the final presentation slides~~
- [x] ~~Prepare the live demonstration of the main features (register, log in, search and book parking)~~
- [x] ~~Assign speaking roles and rehearse the presentation~~
- [x] ~~Deliver the presentation and demonstration and record stakeholder feedback~~

## Day 3: Deployment and Closure

**User Story:** As a project manager, I want to deploy the final version and officially close the project so that the ParkLink application is delivered and all project activities are completed.

**Tasks**

- [x] ~~Prepare the application for deployment and run a final check~~
- [x] ~~Deploy the application and confirm it works in the live environment~~
- [x] ~~Hand over the final code, documentation and project files~~
- [x] ~~Hold the project closure meeting and sign off~~